Activity 4

Take some crushed ice cubes in a beaker and note down the temperature using thermometer. It will be 0 ^{0}C. Now heat the ice in the beaker. You can observe that ice is melting to form water. Record the temperature at regular intervals and it will remain at 0 ^{0}C until whole ice is converted to liquid. Now heat the beaker again and record the temperature. You can notice that the temperature will rise up to 100^{0}C and it will retain the same even after continuous heating until the whole mass of water in the beaker is vaporized.

In the above activity, temperature is constant at 0°C until entire ice is converted into liquid and again constant at 100°C until all the ice is converted into vapor. Why? It is because, when a substance changes from one state to another, a considerable amount of heat energy is absorbed or liberated. This energy is called latent heat. Thus, latent heat is the amount of heat energy absorbed or released by a substance during a change in its physical states without any change in its temperature.

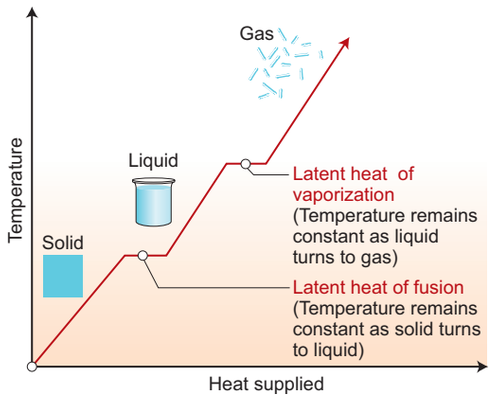


Figure 7.10 Latent heat

Heat energy is absorbed by the solid during melting and an equal amount of heat energy is liberated by the liquid during freezing, without any temperature change. It is called latent heat of fusion. In the same manner, heat energy is absorbed by a liquid during vaporization and an equal amount of heat energy is liberated by the vapor during condensation, without any temperature changes. This is called latent heat of vaporization.

Specific latent heat

Latent heat, when expressed per unit mass of a substance, is called specific latent heat. It is denoted by the symbol L. If Q is the amount of heat energy absorbed or liberated by 'm' mass of a substance during its change of phase at a constant temperature, then specific latent heat is given as L = Q/m.

Thus, specific latent heat is the amount of heat energy absorbed or liberated by unit mass of a substance during change of state without causing any change in temperature. The SI unit of specific latent heat is J/kg.

Problem 5

How much heat energy is required to melt 5 kg of ice? (Specific latent heat of ice = 336 Jg^{-1})

Solution:

Given, m = 5 Kg = 5000g, L = 336 Jg^{-1}

Heat energy required = m × L

= 5000 × 336

= 1680000J or 1.68 × 10^6 J

Problem 6

How much boiling water at 100°C is needed to melt 2 kg of ice so that the mixture which is all water is at 0°C?

[Specific heat capacity of water = 4.2 JKg^{-1} and Specific latent heat of ice = 336 Jg^{-1}].

Solution:

Given, mass of ice = 2 kg = 2000 g.

Let 'm' be the mass of boiling water required.

Heat lost = Heat gained.

$m \times c \times \Delta t = m \times L$

$m \times 4.2 \times (100 - 0) = 2000 \times 336$

$$m = \frac{2000 \times 336}{4.2 \times 100}$$

= 1600 g or 1.6 kg.

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